EarthCube Oceanography and Geobiology Environmental 'Omics

ECOGEO

# ECOGEO Workshop 2: Introduction to Env 'Omics

## Amplicon Analysis - mothur

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# mothur Background - 1

- 1 of 2 majorly used pipelines for the analysis of 16S rRNA amplicon sequences – Qiime
  - iTagger (JGI)
- “mothur is currently the most cited bioinformatics tool for analyzing 16S rRNA gene sequences. …[T]he wiki and user forum and [contain] how you can use mothur to process data generated by Sanger, PacBio, IonTorrent, 454, and Illumina (MiSeq/HiSeq).”

# mothur Background - 2

“mothur and QIIME” P. Schloss (2016) - http://blog.mothur.org/2016/01/12/mothur-and-qiime/

In reference to reviewers: [T]he differences that they think are huge are largely cosmetic.”

Many of the same functions are available in both: cleaning sequences, clustering OTUs, determining alpha and beta diversities, generation of BIOM format documents

# Differences

| Mothur | Qiime |
| --- | --- |
| All functions coded in to a single program in C | Written as a wrapper script in Python – individual aspect coded elsewhere |
| Fully open source | Open source – except for components that are not |
| Development progresses mainly through Schloss lab | Lots of collaboration – additional tools, etc |
| Parameter rich – lots of explanations about defaults | Parameter rich – limited explanation about details |
| No build in data viz | Built in data viz |
| OTU clustering – time intensive, heavily supported results | OTU clustering – faster, multiple options, results may vary |
| Classify with naïve Bayesian algorithm | Classify with USEARCH. Requires 2 top matches for assignment. DB order sensitive. |
| Database – default SILVA (Ref123), able to integrate other DBs | Database – default greengenes (2013), able to use other DBs |

# Essential mothur - 1

Remember both mothur and Qiime offer multi-day workshops that delve in to the minutiae and details of these methods – now less than 30 min

Based on the MiSeq Standard Operating Procedure - http://mothur.org/wiki/MiSeq_SOP

# Essential mothur - 2

Reducing sequencing & PCR errors

- make.contigs
  - Schloss lab uses fully overlapping forward and reverse reads to increase fidelity of results
- Remove all sequences that have ambiguities and cannot exist in reality (too long after complete overlap join)

# Essential mothur - 3

Align and trim

- Align sequences to the reference database
- Trim sequences outside of target hypervariable region(s)
- Important to know the position of primers on the fully expanded 16S rRNA alignment!

# Essential mothur - 4

Reduce complexity of sequences

- Combine all identical sequences
- Remove chimeric sequences
- Identify sequences aligned to lineages that “cannot” be in your sample → Chloroplast, Eukaryote, etc.

# Essential mothur - 5

Make OTUs

- Cluster sequences into OTUs – 97% similarity
- Produce OTU table outputs
- Assign OTUs a putative taxonomy

# Essential mothur - 6

Analysis

- Alpha diversity - mean species diversity (no. of species present)
- Broad comparisons between samples and observed richness
  - Rarefaction – counts the number of observed OTUs as the number of sequences analyzed increases
  - Chao 1 – creates an estimate that increases the weight of the presence of rare OTUs
  - Inverse Simpson – “indication of the richness in a community with uniform evenness that would have the same level of diversity”

# mothur (Hands-On) - 1

```
$ cd /home/c-debi/ecogeo/mothurdir
$ mothur
```

To perform rarefaction:

```
mothur > rarefaction.single(shared=stability.an.shared, calc=sobs, freq=100)
```

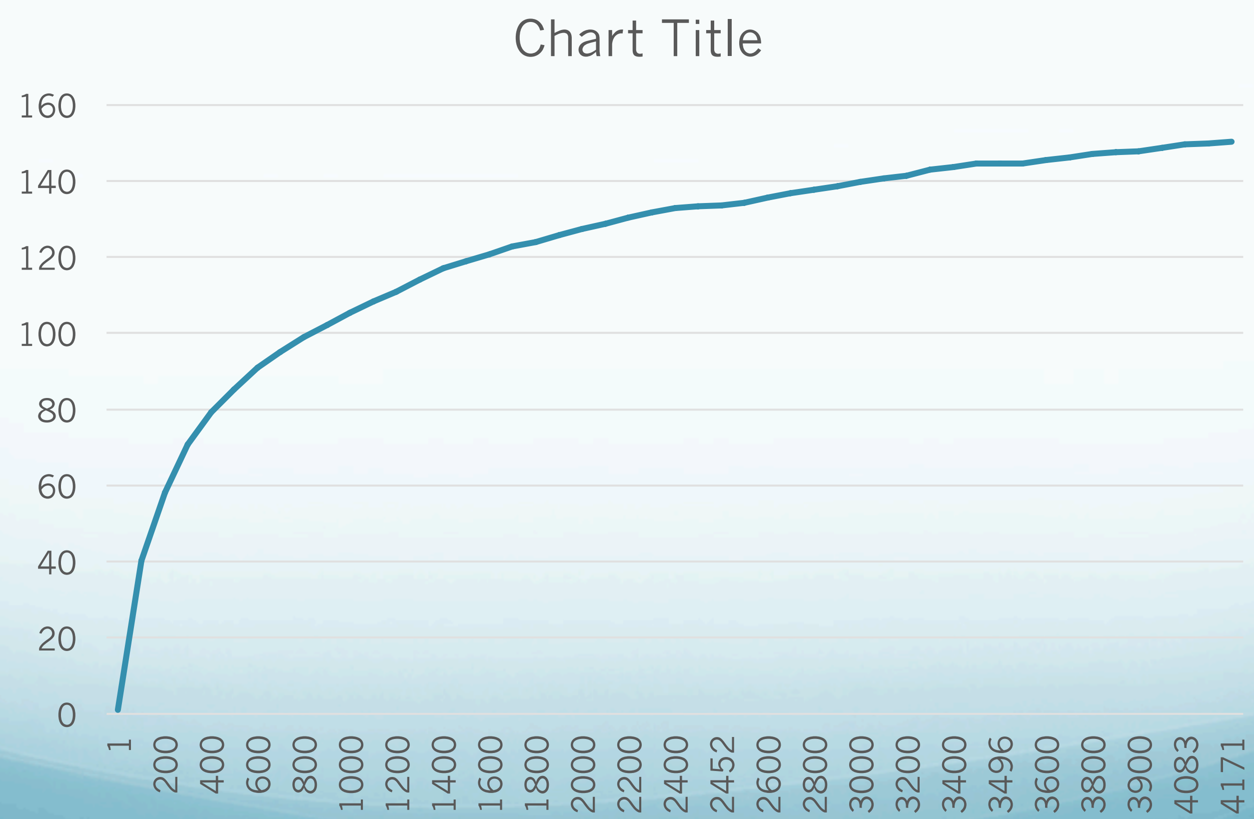
Open a second terminal window, navigate to /mothurdir, less output file

```
numsampled 0.03-F3D0  lci-F3D0   hci-F3D0   0.03-F3D1  lci-F3D1   hci-F3D1   0.03-F3D141     lci-F3D141
     hci-F3D141 0.03-F3D142     lci-F3D142 hci-F3D142 0.03-F3D143     lci-F3D143 hci-F3D143 0.03-F3D144
     lci-F3D144 hci-F3D144 0.03-F3D145     lci-F3D145 hci-F3D145 0.03-F3D146     lci-F3D146 hci-F3D146
     0.03-F3D147     lci-F3D147 hci-F3D147 0.03-F3D148     lci-F3D148 hci-F3D148 0.03-F3D149     lci-F3D149
     hci-F3D149 0.03-F3D150     lci-F3D150 hci-F3D150 0.03-F3D2  lci-F3D2   hci-F3D2   0.03-F3D3  lci-F3D3
     hci-F3D3   0.03-F3Dlci-F3D5 hci-F3D5   0.03-F3D6  lci-F3D6   hci-F3D6   0.03-F3D7  lci-F3D7   hci-F3D7
     0.03-F3D8  lci-F3D8   hci-F3D8   0.03-F3D9  lci-F3D9   hci-F3D9
1    1.0000     1.0000     1.0000     1.0000     1.0000     1.0000     1.0000     1.0000     1.0000
     1.0000     1.0000     1.0000     1.0000     1.0000     1.0000     1.0000     1.0000     1.0000
     1.0000     1.0000     1.0000     1.0000     1.0000     1.0000     1.0000     1.0000     1.0000
     1.0000     1.0000     1.0000     1.0000     1.0000     1.0000     1.0000     1.0000     1.0000
     1.0000     1.0000     1.0000     1.0000     1.0000     1.0000     1.0000     1.0000     1.0000
     1.0000     1.0000     1.0000     1.0000     1.0000     1.0000     1.0000     1.0000     1.0000
     1.0000     1.0000     1.0000
```

# mothur (Hands-On) - 2

```
$ cut -f1,2 stability.an.groups.rarefaction > sample1.rarefaction
```

Creates file with data for rarefaction curve (nano)



# mothur (Hands-On) - 3

## Alpha diversity measures

```
mothur > summary.single(shared=stability.an.shared, calc=nseqs-coverage-sobs-invsimpson, subsample=2441)
```

```
$ less stability.an.groups.ave-std.summary
```

| Sample | No. seqs used | Observed Species | Inv. Simpson | Confidence interval - L | Confidence interval - H |
|---|---|---|---|---|---|
| F3D0 | 2441 | 132.734 | 25.716 | 24.093 | 27.574 |
| F3D1 | 2441 | 127.36 | 34.539 | 32.539 | 37.032 |

# mothur (Hands-On) - 4

What about bar graphs with relative abundance and taxonomy?

```
$ less stability.an.cons.taxonomy
```

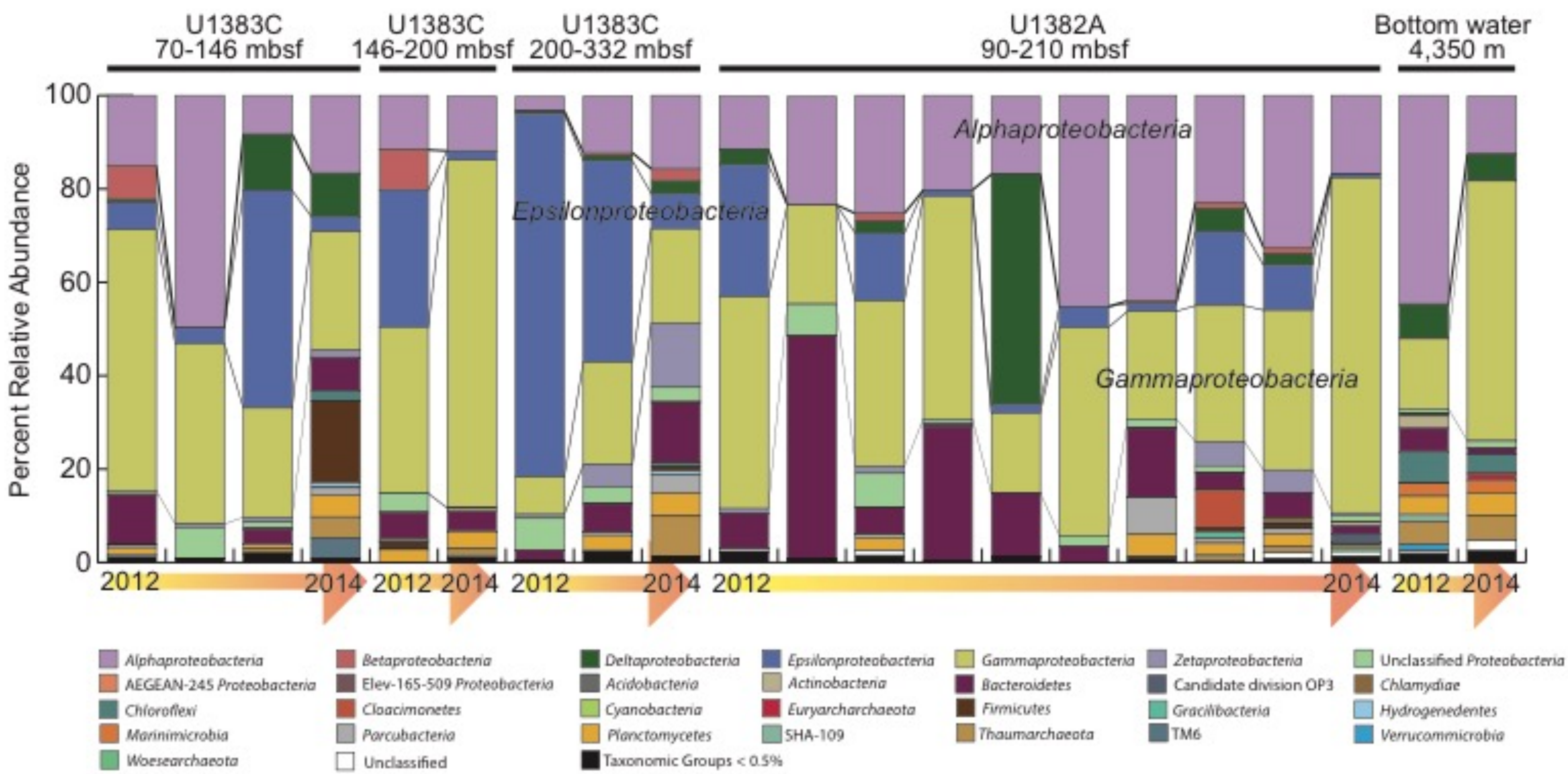
```
OTU  Size Taxonomy
Otu0001   12328

Bacteria(100);"Bacteroidetes"(100);"Bacteroidia"(100);"Bacteroidales"(100);"Porphyromonadaceae"(1
00);"Porphyromonadaceae"_unclassified(100);"Porphyromonadaceae"_unclassified(100);"Porphyromonada
ceae"_unclassified_unclassified(100);
```

```
mothur > sub.sample(shared=stability.an.shared, size=2241)
```

```
$ less stability.an.0.03.subsample.shared
```

```
label Group numOtus      Otu0001      Otu0002      Otu0003      Otu0004      Otu0005      Otu0006      Otu0007
      0.03  F3D0   296    181    116    132    201    164    132
```

# mothur (Hands-On) - 5




Unrelated data

# Essential mothur - 7

Analysis

- Beta diversity - measured differences between communities
- How related are communities to each other?
  - Theta YC (community structure) – considers shared OTUs and relative abundance of OTUs
  - Jaccard (community membership) – compared which OTUs are present in samples

# mothur (Hands-On) - 6

Beta diversity measures

```
mothur > dist.shared(shared=stability.an.shared, calc=thetayc-jclass,
subsample=2241)
```

Turn distance measure output in to a dendrogram

```
mothur > tree.shared(phylip=stability.an.thetayc.0.03.lt.ave.dist)
mothur > quit()
```

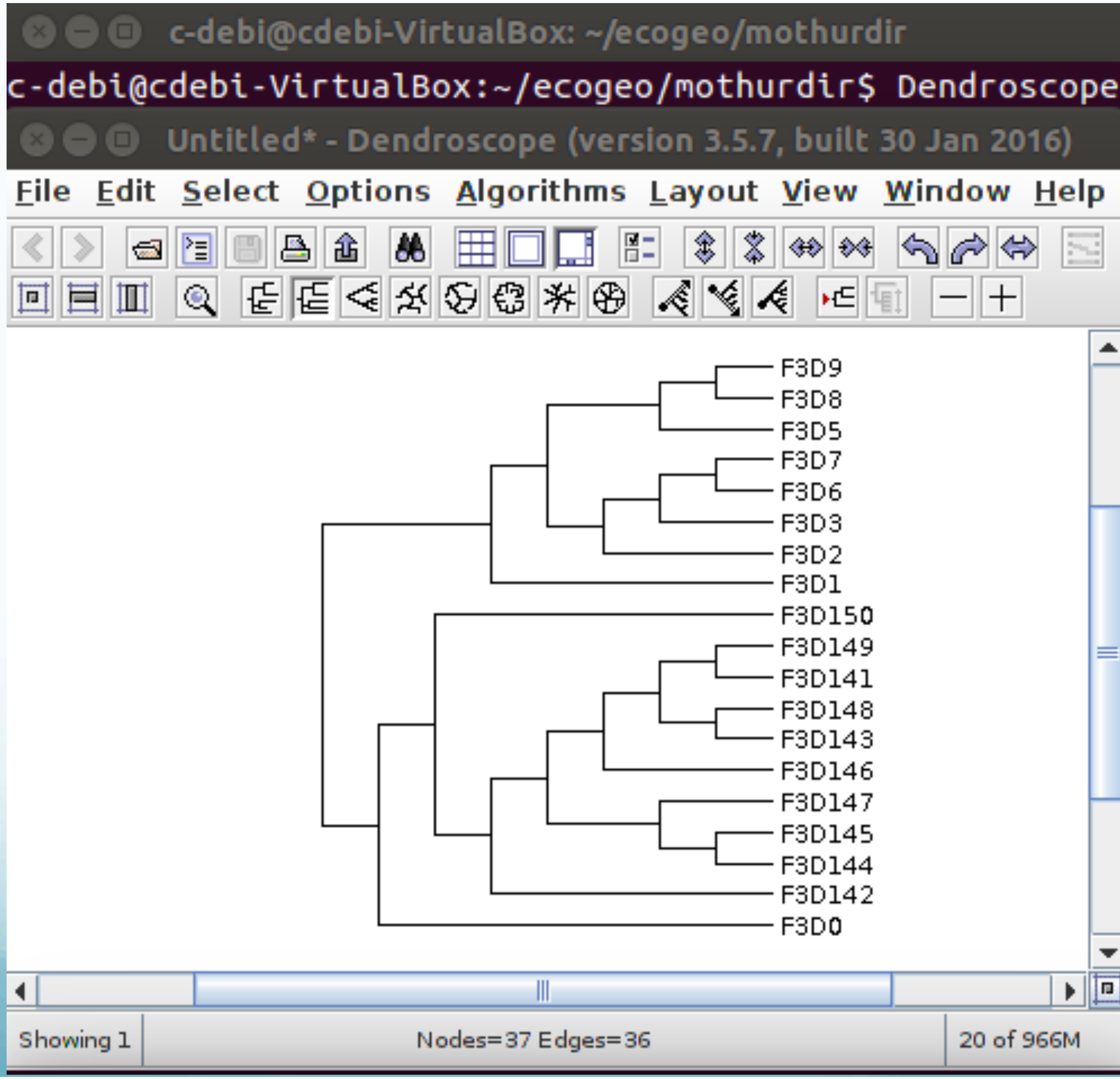
```
$ less stability.an.thetayc.0.03.lt.ave.tre
```

Copy content

```
$ Dendroscope
```

Edit, paste

# mothur (Hands-On) - 7



# Essential mothur - 8

Statistical significance

- Involves classifying your samples in to categories → good experimental design would have classify these samples before seeing results
  - E.g. surface vs deep samples or winter vs. summer
- Generate points to plot Principal Component Analysis or Non-metric Dimensional Scaling plots
- Perform AMOVA statistical tests